BAYESIAN STATISTICS FOR PHD RESEARCHERS · AALBORG UNIVERSITY

Day 1 · Morning
Getting started

Course logistics and your working environment

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Monday 9 November 2026 · 08:45–11:55 · Pontoppidanstræde 101, room 1.001 · luch@math.aau.dk

BEFORE ANYTHING ELSE · 2 MINUTES

How familiar are you with R?

Answer on Mentimeter

Go to **menti.com** and enter code **[code]**. Choose one:

0 · never used it
1 · have run code someone gave me
2 · write small scripts
3 · use it every week

Why we ask

This whole morning is for the technical set-up. Nobody is expected to be fast.

TODAY

Day 1 at a glance

Time	Session	What happens
08:45–09:45	Lecture A	Logistics · R, JupyterLab and help
09:45–11:25	Exercise A (longer today)	Groups · setup · R and JupyterLab practice
11:25–11:55	Peer review A	Short interviews · first plenary
11:55–12:30	Lunch	35 minutes
12:30–13:30	Lecture B	Probability · random variables · marble game · casinos
13:30–15:00	Exercise B	Analyse the timing data
15:00–15:45	Peer review B	Interviews · plenary
15:45–16:00	Daily diary	3 things learned · 3 not yet familiar

COURSE LOGISTICS

Six days, 5 ECTS, three buildings

Day	Date	Room	Group rooms
1	Mon 9 Nov	Pontoppidanstræde 101, 1.001	—
2	Tue 10 Nov	Kroghstræde 3, 4.110	4.130, 4.131
3	Thu 12 Nov	Pontoppidanstræde 101, 1.001	—
4	Fri 13 Nov	Kroghstræde 3, 1.104	4.129, 5.149
5	Mon 16 Nov	Pontoppidanstræde 101, 1.001	—
6	Tue 17 Nov	Niels Jernes Vej 8A, 0.02	0.01

Every day

08:45–16:00

The last 15 minutes are for the daily diary.

Group rooms are for the exercise blocks. Peer review is always in the main room: be back at 11:10 and 15:00.

COURSE LOGISTICS

From your first R command to your own Stan model

DAY 1 · TODAY
R, JupyterLab, probability, uncertainty

DAY 2
Bayesian updating and one-parameter models

DAY 3
Simulation, MCMC and your first Stan model

DAY 4
Multiple parameters and hierarchical models

DAY 5
Gaussian regression and model checking

DAY 6
One GLM and a plan for your own analysis

The goal: by the end you can formulate, fit in Stan, check and explain a Bayesian model for a problem from your own field.

COURSE LOGISTICS

Every day has the same rhythm (from Day 2)

Time	Session
08:45–09:45	Lecture A
09:45–11:10	Exercise A + question set
11:10–11:55	Peer review A
11:55–12:30	Lunch
12:30–13:30	Lecture B
13:30–15:00	Exercise B + question set
15:00–15:45	Peer review B
15:45–16:00	Daily diary

No scheduled breaks

Your group pauses during the exercise block when it suits you.

But fixed return times

Everyone in the room at **11:10** and **15:00**. Interviews need both groups, and any group can be drawn for the plenary.

COURSE LOGISTICS

Ten fixed groups, roles that rotate

10 × 6

- Eight groups of 6, two of 7
- Both cohorts mixed in every group
- Mixed disciplines and R experience
- Same group number (1–10) for all six days

Role	Job in this block
Coder	Drives the shared screen
Checker	Re-runs code, verifies results
Question writers (2)	Draft the group’s question set
Interviewer	Leads the questioning in peer review
Reporter	Notes what was hard; asks it in the plenary

Rotate roles every block, so everyone holds each role at least once. Everyone must be able to explain the group’s work.

COURSE LOGISTICS · THE PEER ROUTINE

You learn by asking each other questions

1 · DURING THE EXERCISES

Write a question set

Three questions with your own worked solutions:

Concept — what does it mean?
Computation — change a value, predict the effect
Pitfall — what could go wrong?

At least one from something your group found hard.

2 · PEER INTERVIEWS · 20 MIN

Interview your partner group

Your partner is the neighbouring group (1 ↔ 2, 3 ↔ 4, …), the same all week. Round 1: odd groups ask. Round 2: swap.

Ask a **named person**; the group may help after the first attempt. Compare with your solution — yours can be wrong too.

3 · PLENARY · 20 MIN

Ask what is still unclear

Two groups are drawn and one interviews the other in front of the class.

Then your **reporter** asks your group's hardest open question. Nothing is uploaded.

COURSE LOGISTICS · THE PEER ROUTINE

The plenary: two groups, drawn at random

1′

Draw

Two group numbers are drawn. The first interviews the second, in front of everyone.

8′

Interview

One or two questions to named members — pick one that caused trouble in your peer round.

10′

Difficult questions

The instructor settles disagreements and corrects wrong answers before we leave the topic.

Any group can be drawn — always have your question set ready. Nobody is skipped: every group appears before anyone is drawn twice.

COURSE LOGISTICS

The daily diary · 15:45–16:00

3

things I learned today

3

things I am not yet familiar with

- Individual, required (part of attendance), **not scored**
- Type your answers into **Mentimeter** (menti.com) — no file, nothing on Moodle
- Your “not yet familiar” lists set the 5-minute follow-up the next morning

COURSE LOGISTICS · GRADING

Half of your grade is earned in this room

50 %

In-class activity

Each **group** scored 0–4 per day by the lecturer. Every present member gets the group's score.

50 %

Final assignment

Individual, after the course, scored out of 100.

Pass

Two conditions

Weighted total **≥ 70 %** and present on **≥ 5 of 6 days**.

What is graded is effort, thinking and participation — not being right. A wrong answer with real reasoning, or a good question about what you don't understand, counts in your favour.

COURSE LOGISTICS · GRADING

What a daily score looks like

Score	The group, that day…
4 Excellent	works and discusses all session; thoughtful question set; everyone contributes in peer review
3 Good	clear effort; reasonable question set; minor gaps
2 Adequate	exercises attempted, little discussion; superficial question set
1 Weak	little visible effort; question set missing or copied; late or passive
0 None	no meaningful participation

COURSE LOGISTICS · FINAL ASSIGNMENT

Seven independent problems

INDIVIDUAL · AFTER THE COURSE

Everyday settings, one problem at a time

Each problem has its own small dataset and does not depend on the others. Most code is provided: you fill in the missing pieces, and every part says exactly what to report.

Scored out of 100: five areas × 20 points.

1 Free throws: updating

2 Help-desk calls: counts, Stan

3 Ice-cream sales: regression

4 Restaurant reviews: pooling

5 Driving test: logistic model

6 Kitchen scale: simulation check

7 Bakery: a decision

Deadline (Danish time, Moodle)	What
13 December 2026, 23:55 — strict	Notebook with outputs, HTML export and your .stan files

COURSE LOGISTICS

If you must miss a day

1

Tell the lecturer in advance — before the day starts, or as early as possible if you fall ill.

2

Submit within a week: that day's in-class core exercises plus Parts A and B of **day-N/extra/**.

3

It is graded 0–4 and counts as your in-class score for that day.

At most **one** missed day. An unannounced absence, no make-up, or a second missed day means the course cannot be passed (except documented exceptional circumstances).

The extra exercises are open to everyone for practice.

WORKING ENVIRONMENT

One conda environment, bayes — today you use two layers

TODAY

JupyterLab

Runs in your browser. Every exercise set is one notebook.

TODAY

R (bayes) kernel

Runs your R code, with ggplot2 for figures.

FROM DAY 3

CmdStanR + Stan

Installed now, used from Day 3 to compile and sample models.

Goal for this morning: **at least two working machines per group** — ideally everyone. You will need your own setup for make-up work and the final assignment.

WORKING ENVIRONMENT · INSTALL ONCE

Four steps to the bayes environment

1 · MINICONDA

Installer from the Miniconda download page. Windows: “Just Me”, not on PATH, then always use **Anaconda Prompt**. Check: **conda --version**.

2 · C++ TOOLS

macOS: **xcode-select --install**
Linux: **sudo apt install build-essential**
Windows: nothing, it comes with step 3.

3 · THE COURSE ENVIRONMENT

```
conda create -n bayes -c conda-forge \
r-base r-cmdstanr r-tidyverse
conda activate bayes
```

4 · JUPYTERLAB + R (BAYES) KERNEL

```
conda install -c conda-forge jupyterlab r-irkernel
Rscript -e "IRkernel::installspec(name='ir-bayes',
displayname='R (bayes)', user=FALSE,
prefix=Sys.getenv('CONDA_PREFIX'))"
```

Check: jupyter kernelspec list shows ir-bayes; in a notebook, **library(cmdstanr)** and **library(tidyverse)** load.

Stuck after 20 minutes? Work on a group partner’s machine now; fix yours later.

WORKING ENVIRONMENT · EVERY TIME YOU WORK

Activate first, then launch JupyterLab

```
conda activate bayes
cd "/path/to/course"
jupyter lab
# Windows (Anaconda Prompt):
# cd /d "C:\path\to\course"
# keep this terminal open; Ctrl+C to stop
```

1. No browser? Copy the link with the **token** from the terminal.
2. Left panel: open **day-1/day1-class.ipynb**.
3. Top right should say **R (bayes)**. If not: Kernel › Change Kernel › R (bayes).
4. Run the first cell with **Shift + Enter**.
5. Finished: File › Save, then Ctrl+C in the terminal.

Symptom	Fix
R code gives Python errors	Kernel › Change Kernel › R (bayes)
CmdStan not found	Quit JupyterLab; conda activate bayes; relaunch

WORKING ENVIRONMENT · NOTEBOOKS

How to do the exercises in a notebook

Code cells

Fill in where it says **___** or **# YOUR CODE**. Output appears below the cell.

Markdown cells

Write your **interpretation** in words. Double-click to edit; Shift + Enter to render.

Before you submit anything

Kernel › **Restart kernel and run all cells**. If it runs top to bottom, it is reproducible.

Keys	Action
Shift + Enter	Run cell, move on
Esc / Enter	Command / edit mode
A · B	New cell above · below
M · Y	Make Markdown · Code
D D	Delete cell
Ctrl/Cmd + S	Save

WORKING ENVIRONMENT · TRY IT NOW

Ten lines of R you will use all day

```
x <- c(9.1, 10.4, 11.8, 8.7) # four guesses, s
err <- x - 10 # errors vs target
mean(err); sd(err); length(err)
sum(err < 0) # how many short?
mean(abs(err)) # typical size

d <- airquality # a data set built into R
head(d); str(d); summary(d)
table(d$Month)
hist(d$Temp)
```

Read it aloud

<- or **=** “gets” (both assign) · **c()** combine · **d$col** one column · **#** comment

Vectorised

x - 10 subtracts from every element; **err < 0** gives TRUE/FALSE, and **sum()** counts the TRUEs.

WORKING ENVIRONMENT · HELP

Finding help on an R function

Type	When
?mean	You know the function’s name
help("sd")	Same, with quotes
??"standard deviation"	You don’t know the name: search all help pages
args(rbinom)	Just remind me of the arguments
example(mean)	Run the examples on the help page

Reading a help page

Skip to **Examples** at the bottom first. Then **Usage** → **Arguments** → **Value** (what you get back).

Reading an error

Read the message slowly; check spelling and capitals; inspect the object with **str()**. Search the exact message text.

WORKING ENVIRONMENT · AI ASSISTANTS

Use AI as a tutor, not as a ghostwriter

Good uses

- Explain an error message (paste code + error)
- Explain what one line of R does
- Rephrase a help page in plain words
- Suggest how to check your own result

Course rules

- You must be able to **explain** all you submit
- **Run and check**: AI is often confidently wrong
- No confidential or unpublished data in prompts
- Your question set comes from **your** difficulties

A PROMPT THAT HELPS YOU LEARN

“I’m learning R in a Jupyter notebook. This code gives the error below. Explain what the error means and how I could have found the cause myself — don’t just give me fixed code.”

EXERCISE A · HELP WARM-UP · ABOUT 10 MINUTES

Find each answer in R's help — not by guessing

H1 · ?mean

What does the argument **trim** do? Predict, then run **mean(c(1, 2, 3, 100), trim = 0.25)**.

H2 · ??"standard deviation"

Which function computes a standard deviation? Does it divide by **n** or **n − 1**? Where on the help page does it say?

H3 · args(rbinom) and ?dbinom

Which three arguments does **rbinom** need? On the same help page, which function gives P(X ≤ x)?

H4 · example(mean)

Run it. Then: why does **mean(c(1, NA, 3))** return NA, and which argument fixes it?

H5 · READ AN ERROR

Run **mean("5")**. What does the warning say, and why? Ask an AI assistant to explain it — then check its answer against **?mean**.

EXERCISE A · 09:45–11:25

Your to-do list, in order

1 Find your group (number on the board). Agree today's roles: coder, checker, 2 question writers, interviewer, reporter.

2 **conda activate bayes**, start JupyterLab. Get **at least two machines** working; help each other.

3 Open **day-1/day-1-exercises-morning.ipynb**, select the **R (bayes)** kernel, run the first cell.

4 Everyone: type and run the **ten lines of R** from the lecture in a new cell.

5 Try the notebook basics: add a Markdown cell, then **Restart kernel and run all cells**.

6 Do the **help warm-up** (previous slide) and the R refresher; use **?function** before asking.

7 Question writers: **two questions** about R or JupyterLab, with solutions. Checker reviews.

8 Reporter: note what your group finds hardest, to ask in the plenary. Take a break when it suits you.

Everyone back in the room at 11:25 for Peer review A.

PEER REVIEW A · 11:25–11:55 · LEARNING THE ROUTINE

Ask your partner group

YOUR PARTNER GROUP · THE SAME EVERY SESSION

1 ↔ 2 · 3 ↔ 4 · 5 ↔ 6

7 ↔ 8 · 9 ↔ 10

TWO ROUNDS · NOBODY MOVES

Round 1 (about 6 min): the odd group asks the even group.

Round 2 (about 6 min): swap.

Turn your chairs to the neighbouring table.

Time	Step
11:25–11:37	Two short interview rounds — one question each, to a named person
11:37–11:55	First plenary: two groups drawn, then your open questions

EXERCISE A RUNS UNTIL 11:25

Back at 11:25. Everyone.

Bring your two questions and their solutions.